■ AI Project Mentor — Fix Summary Report

Date & Time: 2026-11-03 13:49:00
Session Goal / Focus: Strict UB-Mode refactoring: remove duplication, simplify state, and improve readability
Total Fixes Applied: 3

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1. Deduplicate target file syntax verification

Target File:	src/main.py
Category:	improvement
Why / ROI:	Eliminates 12 lines of duplicated compilation logic and prevents inconsistent syntax validation between helper and executor.

Before (Original Code):

```
syntax_error = None
if target_path.suffix == ".py":
try:
py_compile.compile(target_path, doraise=True)
except py_compile.PyCompileError as e:
syntax_error = str(e)
elif target_path.suffix == ".json":
try:
json.loads(target_path.read_text(encoding="utf-8"))
except Exception as e:
syntax_error = str(e)
```

After (Applied Fix):

```
syntax_error = check_file_syntax(target_path)
```

2. Simplify database schema migration columns

Target File:	src/database.py
Category:	improvement
Why / ROI:	Removes 24 lines of repetitive try/except migration statements and centralizes schema evolution in one readable definition list.

Before (Original Code):

```
try:
await db.execute("ALTER TABLE tasks ADD COLUMN session_id TEXT DEFAULT ''")
except Exception:
pass

try:
await db.execute("ALTER TABLE project_state ADD COLUMN current_session_id TEXT DEFAULT ''")
except Exception:
pass
try:
await db.execute("ALTER TABLE tasks ADD COLUMN expected_profit TEXT DEFAULT ''")
except Exception:
pass
try:
await db.execute("ALTER TABLE tasks ADD COLUMN search_block TEXT DEFAULT ''")
except Exception:
pass
try:
await db.execute("ALTER TABLE tasks ADD COLUMN dependent_patches TEXT DEFAULT '[]'")
except Exception:
pass
await db.commit()
```

After (Applied Fix):

```
migrations = [
("tasks", "session_id TEXT DEFAULT ''"),
("project_state", "current_session_id TEXT DEFAULT ''"),
("tasks", "expected_profit TEXT DEFAULT ''"),
("tasks", "search_block TEXT DEFAULT ''"),
("tasks", "dependent_patches TEXT DEFAULT '[]'"),
]
for table, col in migrations:
try:
await db.execute(f"ALTER TABLE {table} ADD COLUMN {col}")
except Exception:
pass
await db.commit()
```

3. Deduplicate state upsert logic

Target File:	src/database.py
Category:	improvement
Why / ROI:	Consolidates project_state column updates into a single idempotent helper method, reducing SQL code duplication across the database module.

Before (Original Code):

```
async def update_task_status(self, task_id: int, status: str):
"""Updates task status and resets ignored_count when user interacts."""
async with self._get_db() as db:
await db.execute("UPDATE tasks SET status = ? WHERE id = ?", (status, task_id))
await db.execute("UPDATE project_state SET ignored_count = 0 WHERE id = 1")
await db.commit()
```

After (Applied Fix):

```
async def update_task_status(self, task_id: int, status: str):
"""Updates task status and resets ignored_count when user interacts."""
async with self._get_db() as db:
await db.execute("UPDATE tasks SET status = ? WHERE id = ?", (status, task_id))
await db.commit()
await self.reset_ignored_count()
```